# Exact Synthesis of Two-Qubit Clifford+CS Circuits with a Near-Minimal Number of K and CS Gates

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We give a method for the decomposition of an arbitrary two-qubit Clifford+CS operator (equivalently a unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$, the Gaussian integers with dyadic denominators), into a Clifford+CS circuit. Given unitary $U$, let $k$ be the least natural number such that entries of $(1+i)^k U$ are all Gaussian integers, then this method decomposes $U$, upto a global phase, into a circuit of length at most $20k + 9$, using at most $2k$ scaled H gates and at most $2k + 1$ Controlled-S gates, without using any ancilla. Moreover, the number of scaled H gate is at most 2 times worse than optimal, the number of *CS* gate is at most 4 times worse than optimal. This method is similar to the work of Russell [16] for the Clifford+$T$ gate set but with more detailed analysis, where the ring in use is $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ which contains $\mathbb{Z}[\frac{1}{2}, i]$.

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## I. INTRODUCTION

In quantum computation, the decomposition of unitary operators into gates from a fixed universal set is an important problem. In this paper we focus on the exact synthesis 2-qubit Clifford+Controlled-S. Previous work on synthesis of Clifford+CS operators or Clifford+T operators mainly focus on minimizing the use of CS gate or T gate since they are more expensive than Clifford operators in certain scenerio. But works on T gate optimization [1, 3, 6, 7, 11, 13] suggested that the H gate is an "obstruction" to their T-optimization method. Quoting from [1] summarizes the reason: "It has been shown that T-depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly".

We propose a method that produces circuits with near-minimal number of both the scaled H gate and CS gate, hence alleviate such "obstruction".

### A. Background

Consider the following unitary operators:

$$i, \quad K = e^{-i\pi/4} H = \frac{1}{1+i} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix},$$

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

Here, $i$ is a scalar, namely the usual complex unit. $H$ is the Hadamard gate, and $K$ is a scaled version of the Hadamard gate. $S$ is sometimes called the *phase gate*, and $CZ$ is the controlled-$Z$ gate. When closed under multiplication, identities, and tensor products, these operators generate the *Clifford group* (possibly up to scalars, depending which scalars are included in the Clifford group — for our purposes, the scalars $\pm 1$ and $\pm i$ are sufficient).

Every operator $U$ obtained in this way is of size $2^n \times 2^n$ for some natural number $n$, and as usual, we say that $U$ is an operator on $n$ qubits. We write $\mathcal{CLI}(n)$ for the group of $n$-qubit Clifford operators (just $\mathcal{CLI}$ if $n = 2$). It is well-known that this group is finite for any given $n$ (see, e.g., [17]), and therefore not universal for quantum computing. We can obtain a universal gate set by adding the controlled phase gate

$$CS = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}.$$

The resulting operators are called the Clifford+$CS$ operators. Note that $CZ = CS^2$. The set of 2-qubit Clifford+$CS$ operators is denoted by $\mathcal{CCS}$. The set of 2-qubit Clifford+$CS$ circuits (a circuit is a finite sequence of gates) is denoted by $\mathcal{CIR}$. Each circuit $D \in \mathcal{CIR}$ implements a unique operator $[D] \in \mathcal{CCS}$.

We use the following notations: we write $K_0 = K \otimes I, K_1 = I \otimes K$, and similarly for $S_0$ and $S_1$ where $I$ is the $2 \times 2$ identity operator. We identify the scalar $i$ with the $4 \times 4$-matrix $i\,(I \otimes I)$. In the notation for controlled gates, we use the convention that the target is the last index. For example, $CX_{01}$ is a controlled $X$-gate with control qubit 0 and target qubit 1, also shorted as $CX$. $CX_{10}$ is shorted as $XC$. The controlled $S$- and $Z$-gates are symmetric, in the sense that $CS_{jk} = CS_{kj}$ and $CZ_{jk} = CZ_{kj}$, and therefore the order of indices does not matter for them, so we just write $CS$ and $CZ$. We also use the $Z$-gate, the $X$-gate, the controlled $K$-gate $CK = I \oplus K$, and the swap gate, denoted by $Ex$. They are related to the "primitive" gates introduced so far by $Z = S^2$, $X = KZKi$, $CX = K_1 CZ K_1 \cdot i$, $Ex = CX\,XC\,CX$, and

$$CK = CZ \cdot Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

Similar to $CX$ gate, $CK_{01}$ is shorted as $CK$ and $CK_{10}$ is shorted as $KC$. By Equation 1, $CK \in \mathcal{CCS}$.

### B. Related work

There are many works on quantum circuit synthsis. Here we only list a few of the close related ones. Giles and Selinger first presented an algorithm for the exact synthesis of $n$-qubit operators with entries in the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ using a method involving decomposition into 1- and 2-level matrices, in which one ancilla is sufficient [2]. If the matrix entries are of the form $\frac{a+b\sqrt{2}+i(c+d\sqrt{2})}{\sqrt{2}^k}$, then this algorithm achieves an upper bound of $O(3^{2^n} nk)$ gates, which is far from optimal.

The problem with this algorithm was that the exponent in the denominator of the remaining entries of the matrix may increase after applying the 1- and 2-level matrices, and this increases the number of operations needed to reduce the matrix. Kliuchnikov then introduced an algorithm which achieved an upper bound of $O(4^n nk)$ gates, using a different method requiring at most two ancillas [3].

| $n$ | # $T$ gates | Time complexity | # ancilla | Ref. |
|---|---|---|---|---|
| 1 | $\mathcal{T}(U)$ | $O(\mathcal{T}(U))$ | 0 | [7] |
| 2 | $\leqslant 10\mathcal{T}(U)$ | $O(\mathcal{T}(U))$ | 0 | this paper |
| $\geqslant 1$ | $O\left(3^{2^n} nk\right)$ | $O\left(\text{poly}\left(2^n, 3^{2^n} nk\right)\right)$ | 1 | [15] |
| $\geqslant 1$ | $O\left(4^n nk\right)$ | $O\left(\text{poly}\left(2^n, 4^n nk\right)\right)$ | 2 | [11] |
| $\geqslant 1$ | $\mathcal{T}(U)$ | $O\left(2^{n\mathcal{T}(U)/2}\right)$ | 0 | [7,16] |

TABLE I. Summary of exact synthesis algorithms. Here $n$ is the number of qubits, $\mathcal{T}(U)$ is the $T$ count, and $k$ is the denominator exponent of the input unitary.

With the revelation that this efficient bound was possible, this paper presents an algorithm that achieves an efficient bound for the 1- and 2-qubit cases while using the original method of decomposition into 1- and 2-level operators in which one ancilla is sufficient.

## II. SOME ALGEBRA

The concepts of this section are similar to those used in [2, 8–10, 14, 15], except in some cases the underlying ring is different.

### A. Rings

As usual, we write $\mathbb{N} = \{0, 1, \ldots\}$ for the set of natural numbers. We also write $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ to denote the rings of integers, real numbers, and complex numbers, respectively. We write $z^\dagger$ for the complex conjugate of a complex number. Recall that in any ring, a *unit* is an invertible element.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. It is a Euclidean domain[4], and therefore, division with remainder, greatest common divisors, divisibility, and the concept of a prime (i.e. a non-unit that cannot be written as the product of two non-units) all can be used in $\mathbb{Z}[i]$. Let $\gamma = 1 + i$. It is well-known that $\gamma$ is a Gaussian prime [12]. Note that $\gamma$ is a divisor of 2, and $\gamma^2$ and 2 divide each other. Using Euclidean division, it is further easy to check that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1+i, 1-i, 2\}.$$

Note that the above is a slight abuse of notation. Normally, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal $(\gamma)$ whose elements are residues nomrally denoted by $x \bmod (\gamma)$ for any $x \in \mathbb{Z}[i]$. Here we omit the mod $(\gamma)$ for simlicity. Note that $\{0, 1\}$ etc. are complete

sets of residue classes and also the reminders from the Euclidean division of $x$ divided by $\gamma$.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. For every $t \in \mathbb{D}[i]$, there exists some natural number $k$ such that $\gamma^k t \in \mathbb{Z}[i]$. This motivates the following definition.

**Definition II.1.** Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for $t$ if $\gamma^k t \in \mathbb{Z}[i]$. The least such $k$ is called the *least denominator exponent* of $t$, denoted by $\mathrm{lde}_\gamma(t)$. Equivalently, the least denominator exponent of $t$ is the smallest $k \in \mathbb{N}$ such that $t$ can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that $k$ is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least $k$ that is a denominator exponent for all of its entries.

**Remark II.2.** By enumerating all 2-qubit Clifford operators, we have that the lde of a 2-qubit Clifford operator is at most 2.

### B. Residue ring

It is well-known that

$$\mathbb{Z}[i]/(\gamma^n) = \left\{\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0,1\}\right\}.$$

We give an elmentary proof by induction on $n$. Note that similar proof is used for proving the case for $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ by Clement[5]. When $n = 1$, the claim obviously holds. Suppose this is true for $n$, i.e., for any $x \in \mathbb{Z}[i]$, $x$ is congruent to some $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$, that is

$$x = \sum_{i=0}^{n-1} b_i\gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0,1\}$, substituting $k$ in and some rewrite gives

$$x = \sum_{i=0}^{n} b_i\gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i\gamma^i \bmod (\gamma^{n+1}).$$

For simplicity, we omit the $\bmod (\gamma^n)$ in $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$. Since $b_i \in \{0,1\}$, we write binary string $(b_0b_1...)_2$ for the summation, and the parenthesis and subscript $_2$ can be omitted if we can infer that $b_0b_1...$ are residues in binary format from the context.

Moreover, residue classes are distinct, i.e. if $b_0b_1...b_{n-1}$ and $b'_0b'_1...b'_{n-1}$ are equivalent, i.e., $\sum_{i=1}^{n-1} b_i\gamma^i - \sum_{i=1}^{n-1} b'_i\gamma^i = k\gamma^n$ for some $k$, then $b_i = b'_i$. Modding $(\gamma)$ on both side gives $b_0 = b'_0$. Then cancelling $b_0$ and $b'_i$, and dividing both side by $\gamma$ give

$$\sum_{i=1}^{n-1} b_i\gamma^{i-1} - \sum_{i=1}^{n-1} b'_i\gamma^{i-1} = k\gamma^{n-1}$$

Then modding $(\gamma)$ on both side agian gives $b_1 = b'_1$. Repeat this process proves $b_i = b'_i$.

We write $\rho_n(x)$ for the binary residue $x \,(\bmod\ \gamma^n)$. From the proof, we see $\rho_n(x)$ is "stable" w.r.t increase of $n$, i.e., the first $n$ digits of $\rho_{n+1}(x)$ is $\rho_n(x)$. Also $\rho_2$ is "stable" under taking complex conjugation, i.e., $\rho_2(x^\dagger) = \rho_2(x)$

We sometimes also write $x \equiv_{\gamma^n} y$ as an abbreviation for $x \equiv y \bmod \gamma^n$. Modulo $\gamma^3$, we have

| $x$ | $\rho_3(x)$ | $x$ | $\rho_3(x)$ | $x$ | $\rho_3(x)$ | $x$ | $\rho_3(x)$ |
|---|---|---|---|---|---|---|---|
| 1 | 100 | $-1$ | 101 | $i$ | 111 | $-i$ | 110 |
| 0 | 000 | $1+i$ | 010 | $1-i$ | 011 | 2 | 001 |

The $(\bmod\ \gamma^n)$ residue classes form a commutative ring and there is a canonical ring homomorphism from $\mathbb{Z}[i]$. In particular, modulo $\gamma^2$, the addition of the ring coincides with the bitwise exclusive or (denoted by $\oplus$) operation; $(10)_2$ is the ring identity; $(00)_2$ the zero of the ring; multiplication by $(01)_2$ is the right shift operation; by distributivity $11 \cdot b_0b_1 = 10 \cdot b_0b_1 + 01 \cdot b_0b_1 = b_0(b_1 \oplus b_0)$. Note that since $\rho_2(i) = 11$, if $\rho_1(x) = 1$, then either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo $\gamma^3$, the addition is

$$b_0b_1b_2 + b'_0b'_1b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where the $\cdot$ is the multiplication of the ring $\mathbb{Z}[i]/(\gamma)$, equivalently the "and" of two bits. The ring identity is 100, zero is 000. Multiplication of 010 is a right shit by 1 digit and multiplication of 001 is a right shit by 2 digits. The rest of the multiplication table can be obtained using distributativity. The additive inverse is

$$-(b_0b_1b_2) = b_0b_1(b_2 \oplus b_0).$$

By looking at the $\rho_3$ table, we see $\rho_3$ is not "stable" under taking complex conjugation, but instead, if $\rho_3(x) = abc$, then $\rho_3(x^\dagger) = ab\bar{c}$. Since $\rho_3(i) = 111$, if $\rho_1(x) = 1$, then there exists $k \in \{0,1,2,3\}$ such that $\rho_3(i^k x) = 100$.

**Definition II.3.** For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say $x$ is *even*, otherwise *odd*.

For example, $2 + 3i$ and $5 + 2i$ are odd, and $2 + 4i$ and $1 + 3i$ are even. In particular, in our context, by an even Gaussian integer we mean one that is divisible by $\gamma$, not necessarily by 2. However if $x \in \mathbb{Z}$, then $x$ is even (odd) as an integer if and only if $x$ is even (odd) as a Gaussian integer. Therefore, our definition of parity extends the usual one on the integers. The following lemmas are straightforward.

**Lemma II.4.** *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then $\alpha$ is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

**Lemma II.5.** *If $\alpha \in \mathbb{Z}[i]$ is odd, then $\alpha \equiv \pm 1, \pm i \pmod{\gamma^3}$. In other words, $\alpha \equiv i^e \pmod{\gamma^3}$ for some $e \in \{0, 1, 2, 3\}$. Moreover, $\alpha \equiv i^e \pmod{\gamma^2}$ for some $e \in \{0, 1\}$.*

**Lemma II.6.** *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Let $k$ be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$, we define the residue function $\rho_n^k(x) = \rho_n(\gamma^k x)$. We call $\rho_n^k(x)$ the $(n, k)$ residue of $x$. If $k$ is a denominator exponent for a vector or matrix $A$, $\rho_n^k(A)$ is defined entry-wise.

**Lemma II.7.** *For $t \in \mathbb{D}[i]$, if $k = lde_\gamma(t) > 0$, then $\gamma^k t$ is odd.*

**Lemma II.8.** *For any $x \in \mathbb{D}[i]$ with denominator exponent $k$, $\rho_2^k(\bar{x}) = \rho_2^k(x)$, where $\bar{\cdot}$ is the complex conjugation.*

**Lemma II.9.** *For $A$ a matrix with lde $k$, $B$ another matrix of proper size with lde $0$, then $\rho_n^k(AB) = \rho_n^k(A)\rho_n^0(B)$ and $\rho_n^k(BA) = \rho_n^0(B)\rho_n^k(A)$.*

This is because $\rho_n^k(\cdot)$ is a ring homomorphism and matrix multiplication only involves ring operations on matrix entries.

### C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$, then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say $i$ acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$.

Given an $n$ residue $b_0 b_1 ... b_{n-1}$, a right shift of it is an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$. Given an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$, a left shift is $b_0 b_1 ... b_n$. Note that we choose to only allow "well-defined" shifts in the sense no information is lost and no unknown is introduced. Note that the usual right shift is used previously when explaining the multiplication of $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$. We use $LS(x)$ for the left shift of $x$ and $RS$ for the right shift.

**Lemma II.10.** *For $t \in \mathbb{D}[i]$, $k$ a denominator exponent of $t$, we have*

$$\rho_{n+1}^{k+1}(t) = \rho_{n+1}^k(\gamma t) = RS(\rho_n^k(t)) \text{ and}$$

*and if $\gamma^k t$ is even,*

$$\rho_n^{k-1}(t) = \rho_n^k(t/\gamma) = LS(\rho_{n+1}^k(t)).$$

Given a $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we know $\rho_2^k(x) = 1b$ and $\rho_2^k(y) = 1b'$ for some $b, b' \in \{0, 1\}$. So $\rho_2^k(x+y) = 1b + 1b' = 0(b \oplus b')$. Then $\rho_1^{k-1}(x+y) = LS(\rho_2^k(x+y)) = b \oplus b'$. Similarly, $\rho_1^{k-1}(x-y) = LS(\rho_2^k(x-y)) = LS(1b - 1b') = LS((1-1)(b-b')) = b \oplus b'$ (subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$, in other words the additive inverse of $b$ is itself). Note that if $b \oplus b' = 1$, then $\text{lde}(x \pm y)$ decreases by 1, and if $b \oplus b' = 0$, $\text{lde}(x \pm y)$ decreases by at least 2 if $k \geq 2$. Since dividing $t$ by $\gamma$ obviously increases $\text{lde}(t)$ by 1, we have for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 10b$ and $\rho_3^k(y) = 10b'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

So we have $\text{lde}(K \begin{bmatrix} x \\ y \end{bmatrix}) = k - 1$ and one entry of it has lde $k-1$ and the other at most $k-2$ if $k > 2$.

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1') \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \end{bmatrix}.$$

## III. SPECIAL UNITARIES

### A. Permutation matrices

Let $a_0a_1a_2a_3$ be a permutation of $\{0, 1, 2, 3\}$. The permutation matrix $P_{a_0a_1a_2a_3}$ is the unitary matrix that maps $e_i$ to $e_{a_i}$, where $\{e_i\}$ is the standard basis. All 24 permutation matrix are Clifford operators and they can be implemented by circuits of lenght at most 3, see Table II.

### B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ (and $\mathbb{Z}[i]$) are of the form

$$\text{diag}(i^a, i^b, i^c, i^d), \text{ where } a, b, c, d \in \{0, 1, 2, 3\}.$$

| $P$ | circuit | $P$ | circuit | $P$ | circuit |
|---|---|---|---|---|---|
| 0123 | identity | 1023 | $X_1 \cdot CX$ | 2103 | $X_0 \cdot XC$ |
| 1203 | $Ex \cdot X_0 \cdot XC$ | 2013 | $Ex \cdot X_1 \cdot CX$ | 0213 | $Ex$ |
| 3210 | $X_1 \cdot X_0$ | 2310 | $X_0 \cdot CX$ | 2130 | $Ex \cdot X_1 \cdot XC$ |
| 3120 | $Ex \cdot X_1 \cdot X_0$ | 1320 | $Ex \cdot X_0 \cdot CX$ | 1230 | $X_1 \cdot XC$ |
| 3012 | $XC \cdot X_1$ | 0312 | $Ex \cdot XC$ | 0132 | $CX$ |
| 3102 | $Ex \cdot CX \cdot X_0$ | 1302 | $Ex \cdot X_0$ | 1032 | $X_1$ |
| 3021 | $Ex \cdot XC \cdot X_1$ | 0321 | $XC$ | 0231 | $Ex \cdot CX$ |
| 3201 | $CX \cdot X_0$ | 2301 | $X_0$ | 2031 | $Ex \cdot X_1$ |

TABLE II. Circuit implementation of 24 permutation matrices.

They can all be implemented by Clifford+CS circuits with length of at most 6 upto a global phase $i^k$, using at most 1 $CS$ gate. This because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k, b_i \in \{0,1\}, k \in \{0,1,2,3\}.$$

are all distinct and diagonal, and it is of cardinality $4^4$ which equals to the cardinality of the set of diagonal unitary matices.

### C. Generalized unitary permutation matrices

A generalized unitary permutation matrix with entres in $\mathbb{D}[i]$ is a matrix with the same nonzero pattern as a permutation matrix (there is exactly one nonzero entry in each row and each column), but the nonzero entry can be any number of unit length, i.e, powers of $i$. It is easy to see all $4 \times 4$ generalized permutation matrices are product of a permutation matrix and a diagonal matrix, hence can be implemented by a circuit of lenght at most 9 containing at most 1 CS gate.

## IV. THE EXACT SYNTHESIS ALGORITHM

### A. Main theorems

**Theorem IV.1.** *For any $A \in U(\mathbb{D}[i])$ with lde $k$, there exists a circuit $C$ implementing $A$, using at most $2k$ $K$ gates, at most $2k+1$ $CS$ gates, and at most $9(2k+1)+2k$ gates upto a global phase.*

Note that the proof (given later) is constructive, so it gives an algorithm for computing such circuit.

### B. Optimality

**Definition IV.2.** Let $D$ be a Clifford+CS circuit. Let $[D]$ be the unitary operator it implements. Define

- $\mathrm{rkc}(D) =$ the number of K gate that appears in D.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D) =$ the number of CS gate that appears in D.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D) =$ the number of gate that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ repectively.

Any Clifford+CS circuit $D$ is an alternating sequence of Clifford part $C_i$ and a $CS$ gate

$$D = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all 2-qubit Clifford circuit, we know $\mathrm{lde}(C_i) \leq 2$. If (2) is $CS$-optimal, i.e., $\mathrm{cs}(D) = n$, then $\mathrm{lde}(D) \leq \mathrm{lde}(C_0) + \mathrm{lde}(CS) + \mathrm{lde}(C_1) \ldots \mathrm{lde}(C_n) \leq 2(n+1)$. We have the relation

$$\mathrm{lde}(D) \leq 2(\mathrm{cs}(D) + 1).$$

On the other hand, our synthesis algorithm output at most $2 \cdot \mathrm{lde}(D)$ K gates and $2 \cdot \mathrm{lde}(D) + 2$ CS gates. So our our synthesis algorithm is at most 4 times worse than optimal CS-count (experiments show it is 2 times worse, see Section **??**). To acheive the lde, we need at least $\mathrm{lde}(D)$ many K gates, so our algorithm is at most 2 times worse than the optimal K-count.

Clifford+CS circuit $D$ is also an alternating sequence of generalized permutation part $P_i$ and a $K_1$ gate (by replacing all appearences of $K_0$ by $Ex \cdot K_1 \cdot Ex$):

$$D = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

So we have $\mathrm{lde}(D) \leq \mathrm{rkc}(D)$ and then $\mathrm{lde}(D) \leq \mathrm{kc}(D)$. On the other hand, Theorem IV.1 outputs $D'$ such that $\mathrm{rkc}(D') \leq 2\mathrm{lde}(D)$, so we have $\mathrm{kc}(D) \leq 2\mathrm{lde}(D)$.

$$\mathrm{lde}(D) \leq \mathrm{kc}(D) \leq 2\mathrm{lde}(D)$$

If (3) is $K$-optimal, i.e., $\mathrm{kc}(D) = n$, then

$$\mathrm{rcs}(D) \leq \mathrm{kc}(D) + 1 \implies \mathrm{cs}(D) \leq \mathrm{kc}(D) + 1.$$

### C. Lemmas and proofs

We start with a number of lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of $\mathbb{C}$, the usual properties of complex numbers, complex vectors, and complex matrices, such as complex conjugation (denoted by $\bar{\cdot}$), inner product, norm, and conjugate transposition (denoted by $\cdot^\dagger$) are naturally inherited. For example, for a vector $v \in \mathbb{D}[i]^n$, the *norm* of $v$ is $\|v\| = \sqrt{v^\dagger v}$, where $v^\dagger$ is the conjugate transpose of $v$. Note that $\|v\|^2 = v^\dagger v \in \mathbb{D}$, since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly if $v \in \mathbb{Z}[i]$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1.

If $v$ is a vector, the notation $v_j$ refers to its $j$th entry; entries are numbered starting from $j = 0$.

**Lemma IV.3.** *For any $A \in U(\mathbb{D}[i])$ with lde $k$, there exists permutation matrix $L$ and $R$ such that, $\rho_1^k(LAR)$ or $\rho_1^k((LAR)^T)$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

*We also say $A$ has such pattern and write* $\mathrm{pat}(A)$ *for the pattern of $A$. Moreover, $A$ has pattern (i) if and only if $\mathrm{lde}(A) = 0$, i.e., when $A$ is a generalized permutation. $A$ has pattern (iv), (v), and (vi) only if $\mathrm{lde}(A) > 1$.*

Before proving this lemma, we prove serveral useful lemmas.

**Lemma IV.4.** *Let $v$ be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$ then there exists a generalized unitary permutation matrix $Q$ such that $\rho_1^k(Qv)$ equals $(1,1,0,0)^T$ or $(1,1,1,1)^T$. Moreover if $k > 1$ there exists a generalized unitary permutation matrix $Q$ such that $\rho_2^k(Qv)$ equals to $(10,10,01,00)^T$ or $(10,10,10,10)^T$, and if $k = 1$, $\rho_2^1(Qv)$ equals to $(10,10,00,00)^T$.*

*Proof.* Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show $w$ has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\| = |\gamma|^{2k}\|v\| = (\gamma\gamma^\dagger)^k \equiv 0 \,(\mathrm{mod}\ \gamma).$$

Therefore, there are an even number of $j \in \{0, ..., 3\}$ such that $\|w_j\|^2$ is odd. It follows by Lemma II.4 that an even number of $w_j$ are odd.

We can find a permutation matrix $P$ such that $\rho_2^k(Pv) = (1a, 1b, 0c, 0d)^T$ or $(1a, 1b, 1c, 1d)^T$. For the later, by muliplying a diagonal matrix, i.e., multiplying $i$ to the required entries, we can make the second digit all zero, i.e., $(10, 10, 10, 10)^T$.

For the former, by muliplying a diagonal matrix, we can make second digit of the odd entries zero, i.e., $(10, 10, 0c, 0d)^T$. Now applying $K$ to the pair of odd entries we get $(00, 01, 0c, 0d)^T$ or $(01, 00, 0c, 0d)^T$. Note that multiplication of $v$ and a unitary is still a uinit vector, so is applying $K$ to the pair of odd entries. So $(01, 00, 0c, 0d)^T$ or $(00, 01, 0c, 0d)^T$ is a residue of a unit vector with lde $k-1$, then its one-digit $k-1$ residue must of the form $(1, 1, 0, 0)^T$ upto a permutation, that is $c = 1$ and $d = 0$ upto a permutation.

If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ upto a permutation. □

**Lemma IV.5.** *Let $u, v$ be two unit vectors in $\mathbb{D}[i]^4$ with lde $k > 0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized unitary permutation matrix $Q$ such that $\rho_1^k(Q[u|v])$ equals*

$$\begin{bmatrix}1&1\\1&1\\0&0\\0&0\end{bmatrix},\ \begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\1&0\\1&0\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\0&1\\0&1\end{bmatrix},\ or\ \begin{bmatrix}1&1\\1&1\\1&1\\1&1\end{bmatrix}.$$

*Proof.* By Lemma IV.4, there can be only even number of 1s in each column. The only missing case is

$$\begin{bmatrix}1&0\\1&1\\0&1\\0&0\end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(\bar{u}_i v_i) = 1$ contradicts that $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices such that at which the $(1, k)$ residue of $u, v$ is 1 must be even. □

*Proof of Lemma IV.3.* Since the rows and columns of a unitary matrix are unit vectors and all row(column) vectors are perpendicular to each other. Lemma IV.4 and IV.5 forces the pattern to be (ii) - (vi) if $k > 0$. If $k = 0$, the only unitaries with lde 0 are the generalized permutations which has pattern (i). If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ upto a permutation. So residue of such vectors cannot have four 1s. So Case (iv), (v), and (vi) only happens if the lde $> 1$. □

**Lemma IV.6.** *For any $A \in U(\mathbb{D}[i])$ with lde $k > 0$, there exist generalized permutations $L, R, L', R'$ such that*

- *$\mathrm{lde}(K_1 LAR) \le k - 1$;*
- *or $\mathrm{lde}(LARK_1) \le k - 1$;*
- *or $\mathrm{lde}(L'K_1 LARR'K_1) \le k - 1$;*
- *or $\mathrm{lde}(K_1 L'K_1 LARR') \le k - 1$;*
- *or $\mathrm{lde}(L'CKLAR) \le k - 1$ (only happens when $\mathrm{lde}(A) = 1$).*

Note that each $\le$ above is actually $=$, but we don't need this. The proof proceeds by a case distinction over $\rho_1^k(A)$. Note that requiring $k > 0$ eliminates Case i. So start with Case ii.

### 1. Case ii

If $k > 1$, by Lemma IV.4, upto multilying generalized permutations on the left and right, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&01&00\\10&1a&0b&0\bar{b}\\01&0c&0d&0e\\00&0\bar{c}&0f&0g\end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and $b, c, d, e, f, g$ are indeterminates. We claim $a = 0, b = c = 1$. Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the second and third row gives

$$01 + 0c + 00 + 00 = 00 \Rightarrow c = 1.$$

Similarly, we have $b = 1$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix}.$$

Moreover, since $k > 1$ and the third and fourth row (column) is of lde $k - 1 > 0$, $d = f$, $d = e$, $e = g$, i.e., $d = e = f = g$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0d \\ 00 & 00 & 0d & 0d \end{bmatrix}.$$

Applying $K_1 \cdot S_1$ on the left gives Case iv.

If $k = 1$, upto multilying generalized permutations on the left and right, pat$(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix},$$

As before, we have $a = 0$. Apply $X_0 \cdot CK \cdot X_0$ on the left decreases the lde.

### 2. Case iii

If $k > 1$, by Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix},$$

Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Similarly $d = 0$. Taking inner product of the second and third row gives

$$01 + 0c + 0b + 0\bar{b} = 00 \Rightarrow c = 0.$$

Similary, we have $b = 0$. pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying $K_1 S_1$ on the left bring us to Case vi.

If $k = 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, we have $a = 0, d = 0$. Applying $K_1$ on the left decrease the lde.

### 3. Case iv

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Similarly we have $b = c = 0$. Since the third row is of lde $k - 1 > 0$, either $d = e = f = 1$ or one of $d, e, f$ is 1 and the rest are 0. For the later, upto a permutation of columns, we may assume $d = 1, e = f = 0$. For both cases, right multiplication of $K_1$ decrease the lde.

### 4. Case v

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix},$$

We claim $a = 0, b = c = e = 1$. Taking inner product of the first two row gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the first and third row gives

$$10 + 1b + 01 + 00 = 00 \Rightarrow b = 1.$$

Taking inner product of the second and fourth row gives

$$10 + 1c + 00 + 01 = 00 \Rightarrow c = 1.$$

Taking inner product of the second and third column gives

$$01 + 00 + 11 + 1\bar{e} = 00 \Rightarrow e = 1.$$

Similarly $f = 1$. Now pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix},$$

Applying $K_1$ on the left bring us to Case iv.

### 5. *Case vi*

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1x \end{bmatrix},$$

Taking inner product of first two columns

$$a + b + c = 0 \Rightarrow a, b, c \text{ cannot be all 1.}$$

Upto a permutation, we maybe assume $a = 0$. Then $b = c$. Similarly $d = g$.

If $b = c = 1$, and $d = g = 1$, pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1x \end{bmatrix},$$

Taking inner product of the first and third column gives $f = \bar{e}$. Similarly, $h = \bar{e}$ and $x = \bar{f}$. Now pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We prove this pattern cannot happen by looking at its (3,k)-residue. Upto multiplication oof powers of $-1$ on the proper rows and columns, the (3,k)-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11a \\ 101 & 11y & 1ev & 1\bar{e}b \\ 101 & 11z & 1\bar{e}w & 1ec \end{bmatrix}.$$

Unitarity of $A$ and the $k > 1$ requires that the any two rows (columns) are perperdicular to each other. In particular, by taking inner product of the first columns and the second, third column, we have

$$101 + 10x + 11y + 11z = 000 \implies x \oplus y \oplus z = 1$$
$$101 + 11u + 1ev + 1\bar{e}w = 000 \implies u \oplus v \oplus w = 1$$

Taking inner product of the second and thired column gives

$$101 + 11(u \oplus \bar{x}) + (1ev + 01e + 00\bar{y}) + (1\bar{e}w + 01\bar{e} + 00\bar{z})$$
$$= 000 \implies x \oplus y \oplus z \oplus u \oplus v \oplus w = 1, \text{ a contradiction.}$$

If $b = c = 0$, taking inner product of the first and third column gives $e = h$, similarly, $f = x$. So pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Taking inner product of the first and third row gives $d \oplus e \oplus f = 0$, then at leat one of $d, e, f$ is 0, upto a permutation of rows, we may assume $d = 0$. Similar analysis applies if $d = g = 0$. So, for either case, upto permutations of rows and columns pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Note that $e = f$, which is implied by taking inner product of the first and third column. So left multiplication of $K_1$ reduces the lde.

The transitions between cases in summarized in Figure 1.

$$\begin{tikzcd}
\text{ii} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} \arrow[r, "\text{if lde}=1,\ CK\times"] \arrow[d, "K_1\times"'] & \begin{bmatrix} 0&0&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} & \text{iii} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 0&0&1&1 \\ 0&0&1&1 \end{bmatrix} \arrow[l, "\text{if lde}=1,\ K_1\times"'] \arrow[d, "K_1\times"] \\
\text{iv} \begin{bmatrix} 1&1&1&1 \\ 1&1&1&1 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} \arrow[ur, "\times K_1 \text{ or } K_1\times"] & \text{v} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 1&1&1&1 \\ 1&1&1&1 \end{bmatrix} \arrow[l, "K_1\times"] & \text{vi} \begin{bmatrix} 1&1&1&1 \\ 1&1&1&1 \\ 1&1&1&1 \\ 1&1&1&1 \end{bmatrix} \arrow[ul, "K_1\times"]
\end{tikzcd}$$

FIG. 1. Transitions between cases. The $K_1\times$ ($\times K_1$) annotated arrow means if $A$ is of the pattern at the tail, then $K_1LAR$ ($LARK_1$) is of the pattern at the head for appropritate choosen generalized permutations $L$ and $R$. Similarly, for other annotations. Note that $CK$ decomposes to two $K$ gates as in Equation 1.

### D. Complexity

The time complexity is $O(\mathrm{lde}(U))$.

## V. EXPERIMENT

We implemented our algorithm in Haskell and did an comparison with the CS-optimal synthesis algorithm. Circuits are randomly genrated feeded to both algorithm, and output circuits are compared on metrics $\mathrm{rcs}, \mathrm{rkc}, \mathrm{rlen}$.

Statistics show our algorithm alway acheives the optimal CS-count, which suggests that part of our analysis is not tight, and could be improved. This is left to futurn work.

## VI. CONCLUSION AND FUTURE WORK

In conclusion, we proposed an efficient and optimal algorithm for the exact synthesis of Clifford+$CS$ circuits from a given unitary matrix.

The algorithm Li et al. gave outputs a circuit that is at most 10 times worse than optimal in T-count. We believe our method can be used to improve the bound 10.


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